Functional Specification and Overview

Metro Ticket Generating System in ServiceNow | Functional documentation | 11 October 2026

This specification explains the commuter journey, catalog fields, fare display, and QR presentation used in the Skill Wallet metro ticket capstone. It adapts the supplied project documentation format to the ServiceNow implementation.

1 Introduction

This project uses a ServiceNow Personal Developer Instance to present a metro booking form through Service Portal. The configured catalog item is named Book a Ticket.

2 Project Overview

The objective is to give commuters a digital route selection and ticket preview experience. The form gathers station, journey, passenger, and payment mode information. Catalog client scripts display a calculated fare and open a QR presentation widget. The QR address supplied in the Skill Wallet exercise is an example ticket URL for demonstration.

3 Users and Responsibilities

Actor	Responsibility
Passenger	Choose stations and journey options; view fare and QR preview.
Station manager	Review the intended scanning and checking journey in the demo.
Metro operations	Review station data and route coverage.
IT administrator	Maintain catalog item, scripts, widget, and table permissions.

4 Booking Journey

1. Open the Service Portal and find Book a Ticket under Service Catalog > Services.
2. Choose Starting From and Going To from the Metro Station Details reference list.
3. Choose Single Journey or Return Journey and the passenger count.
4. Review the calculated amount shown in the read only fare field.
5. Choose UPI, Card, or Others. The UI policy shows Enter Payment Mode for Others.
6. Use the catalog checkout action to invoke the QRGeneration client script and Metro QR Widget preview.

5 Form Variables

Question	Variable name	Type	Order	Purpose
Starting From	starting_from	Reference	100	Required; references Metro Station Details
Going To	going_to	Reference	200	Required; references Metro Station Details
Type of Jouney	type_of_jouney	Multiple Choice	300	Single or return journey
Number of passengers	number_of_passengers	Numeric Scale	400	Passenger count used by fare script
Amount for Single Journey	amount_for_single_journey	Single Line Text	500	Calculated, read only display
Amount Including Return	amount_including_return	Single Line Text	600	Calculated, read only display
Mode of Payment	mode_of_payment	Multiple Choice	700	UPI, Card, Others
Enter Payment Mode	enter_payment_mode	Single Line Text	800	Shown when Others is selected

The station selectors are required references. The fare fields are read only. The current numeric scale displays 0 through 5; the stored fare script calculates a quote for integer passenger counts from 1 through 4. This document records the configured behavior without changing the item.

6 Business Rules

- The start and destination must be selected before a fare is calculated.
- The fare script uses the ordered 27 station Red Line matrix and a value mapping in the stored script.
- For a return journey, the script treats the fare as twice the one way total.
- A selected payment mode controls the visibility and mandatory state of Enter Payment Mode when Others is chosen.
- The QR widget displays the image URL passed by the onSubmit script through spModal.

7 Acceptance Scenarios

Scenario	Expected visible outcome
Open catalog item	The Book a Ticket form and eight variables appear.
Choose distinct stations and one passenger	The relevant fare field shows a calculated amount.
Choose Others	Enter Payment Mode appears and becomes mandatory.
Invoke QR preview	A modal shows the Metro QR Widget and scannable image.
Review station data	Station reference list contains the configured 27 records.

8 Navigation

Find Book a Ticket in Service Catalog > Services, Metro Station Details in System Definition > Tables, and Metro QR Widget in Service Portal > Widgets. The project README lists the included documentation and source snapshots.